WSU AIRPACT PROJECT UPDATE & ANNUAL REPORT

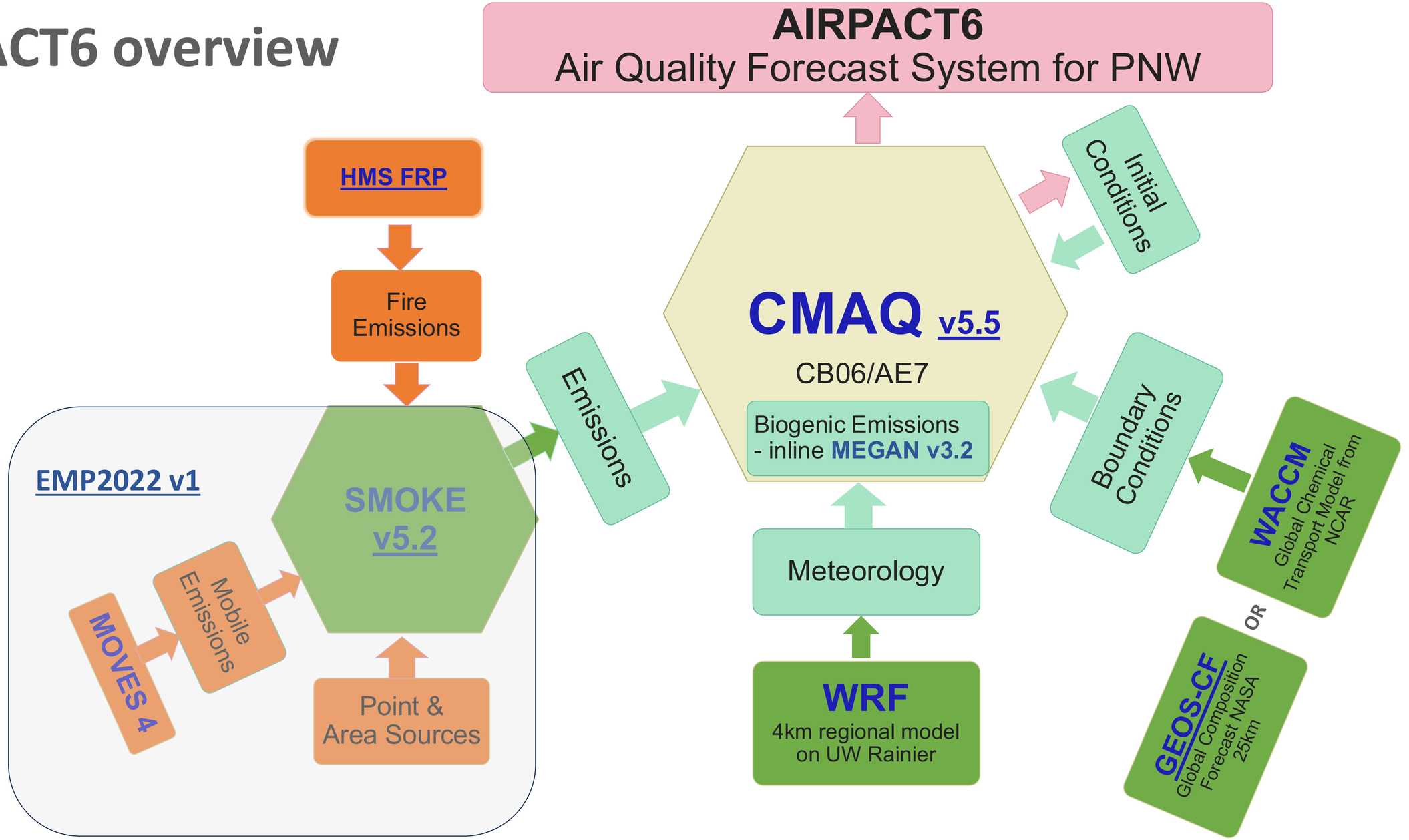
Jun Meng

with contribution from:
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Kimi Smith (IDEQ), Neda Khosravi (ODEQ), Farren Thorpe (WA ECY), Wei Zhang (IDEQ)
and other NW-AIRQUEST members

NW-AIRQUEST Consortium Annual Meeting, Oct 30, 2026

AIRPACT6 is running forecast on new WSU Kamiak HPC

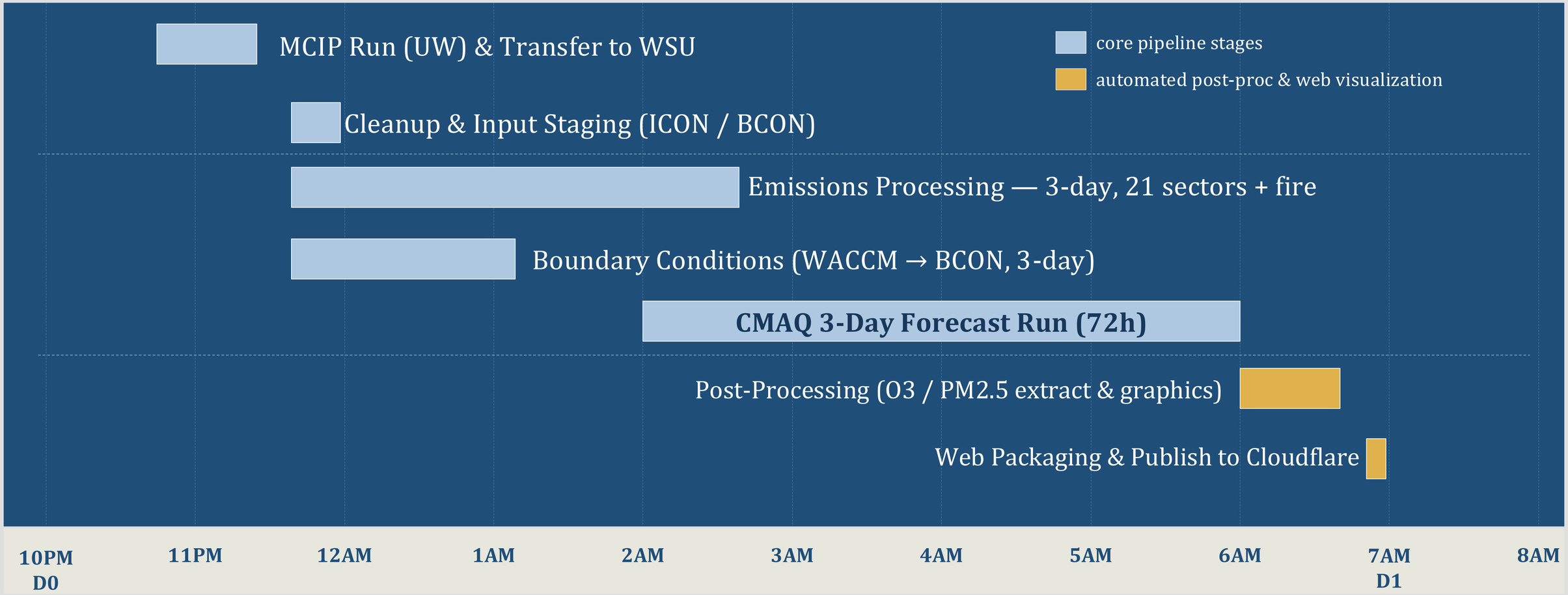
AIRPACT6 overview



AIRPACT-6 OPERATION

AIRPACT5 operation needs around 11 hours; while AIRPACT6 needs around 8 hours

- 3-day forecast (D1–D3): single 72h CMAQ v5.5 run starting from 10:40pm nightly
- automated post-processing & web visualization (Cloudflare for now)
- Operation Schedule (nightly cycle, Pacific Time):



Operation schedule (Pacific Time). Bar lengths are current SLURM time allocations with 3 full nodes — real run-times are less.

Credits to Priom Zarrah

Upgrading to New Systems

CMAQ v5.5

Model

- Upgrade from the AIRPACT-5 CMAQ release
- CB6r5 gas-phase chemistry + AE7 aerosol module
- Updated photochemistry & deposition science

SMOKE v5.2

Emissions processing

- Runs on EPA's 2022v1 modeling platform
- Inline fire, point emissions MEGAN inline biogenic

Model Setup

Chemical Mechanism
CB6R5/AE7

Biogenic Emissions
MEGAN v3 (Inline)

Emission Inventory
EMP 2022v1 (SMOKE v5.2)

Domain
285×258 cells, 4-km, LCC

Boundary Conditions
WACCM

Meteorology
WRF - MCIP

HPC Configuration

3
Compute Nodes

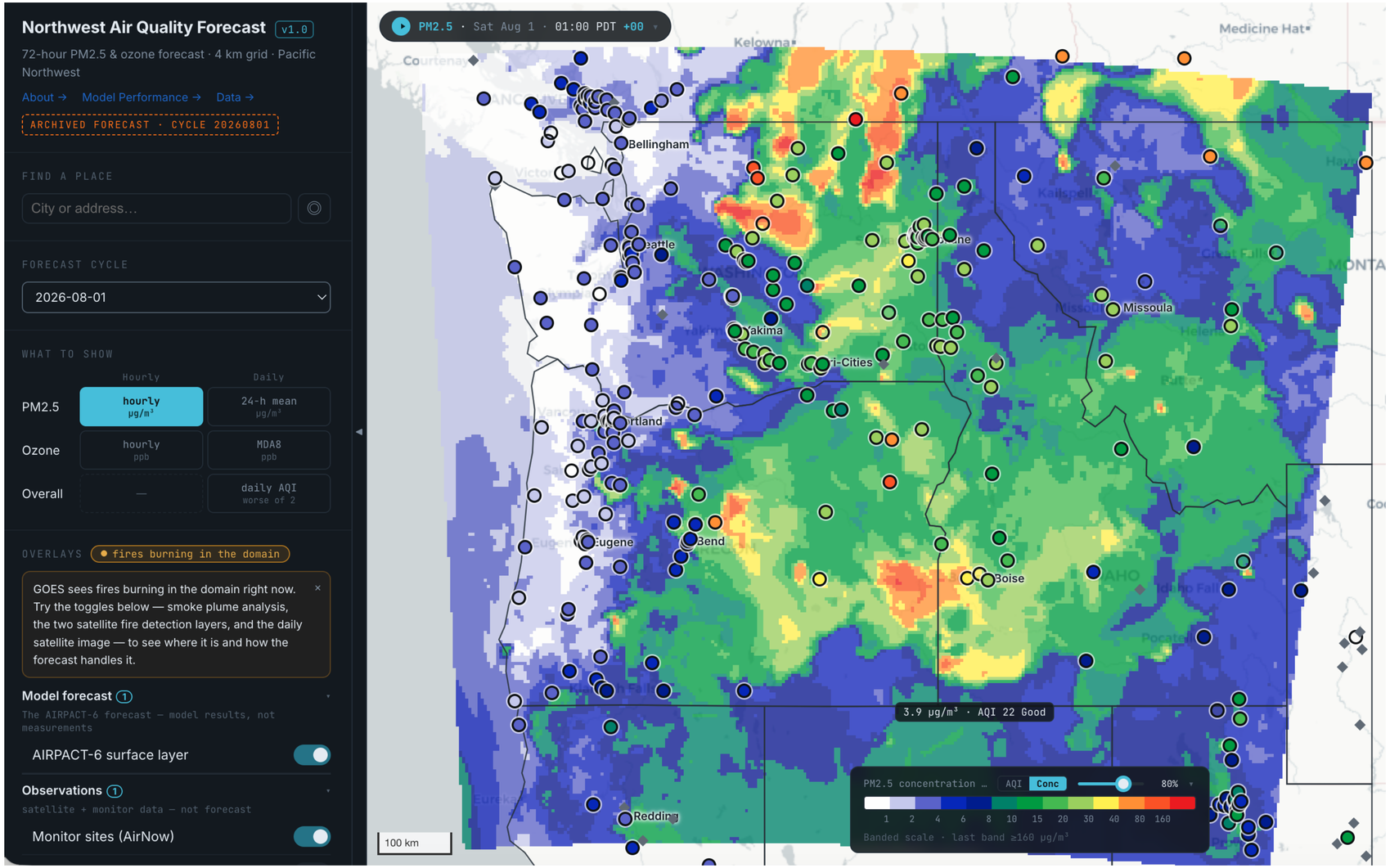
64
CPUs per Node

~2 hrs/day
Wall Time

FY27 Deliverables

AIRPACT6 visualization goes live before the 2026 wildfire season

Interactive Map - https://nw-air-forecast.pages.dev/

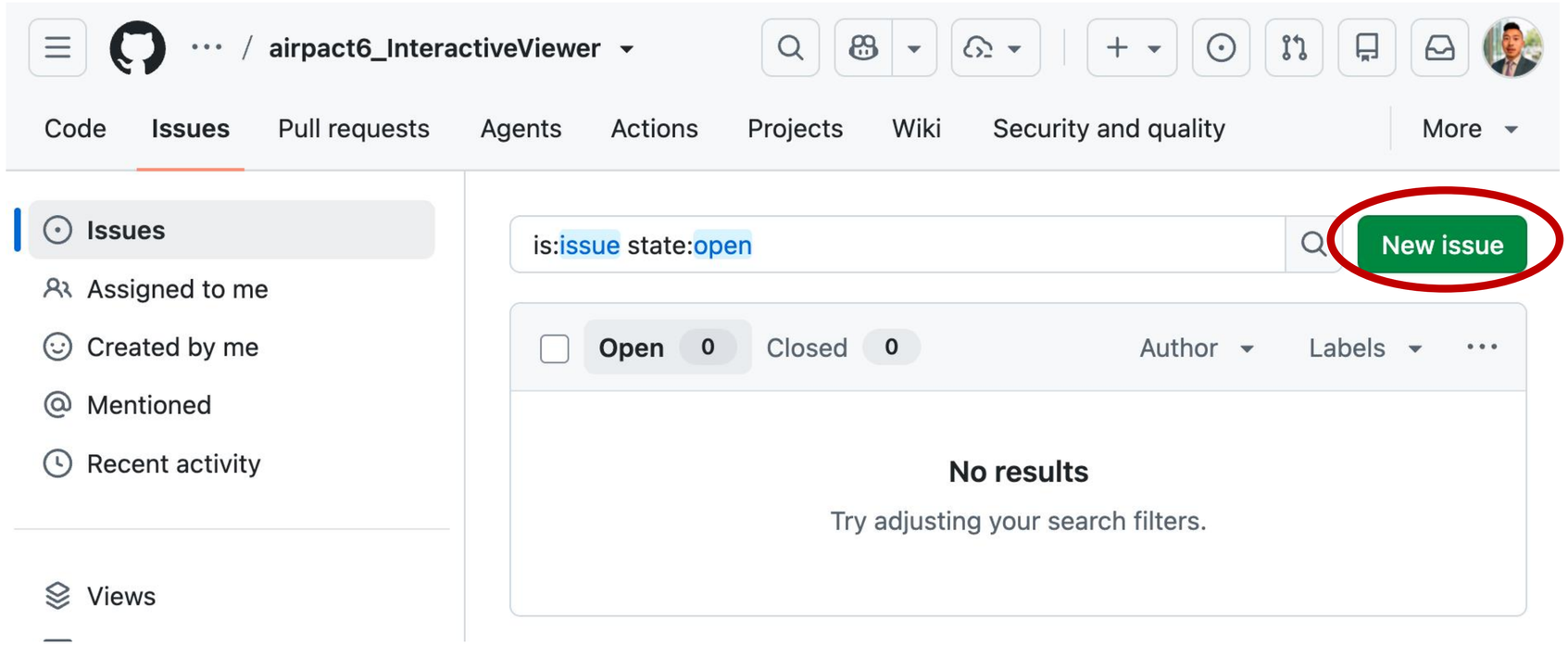


- Currently hosted on Cloudflare – a free cloud web server
- Plan to incorporate it into WSU campus web server – airpact.wsu.edu
- Current features: NRT AirNow overlay, click and query concentration, user-friendly time bar control
- Let us know what features should be changed and what new features should be added

Help us to co-develop this tool

Interactive Map - https://nw-air-forecast.pages.dev/
Github repo - https://github.com/Jun-Meng/airpact6_InteractiveViewer

How you can help:



- Bugs
- New feature suggestions
- Comments

AIRPACT6 model daily data is available online

Data downloads

← back to the map · model performance · about

Everything the interactive map shows is downloadable here for research use: the gridded 72-hour forecast fields, the site's own compact binary format, and the observation-vs-forecast verification statistics. Please see the citation and the as-is disclaimer at the bottom.

Forecast cycle: 2026-10-29 (latest) — init 2026-10-29 01:00 PT · 72 hourly steps · grid 310×270 (EPSG:3857)

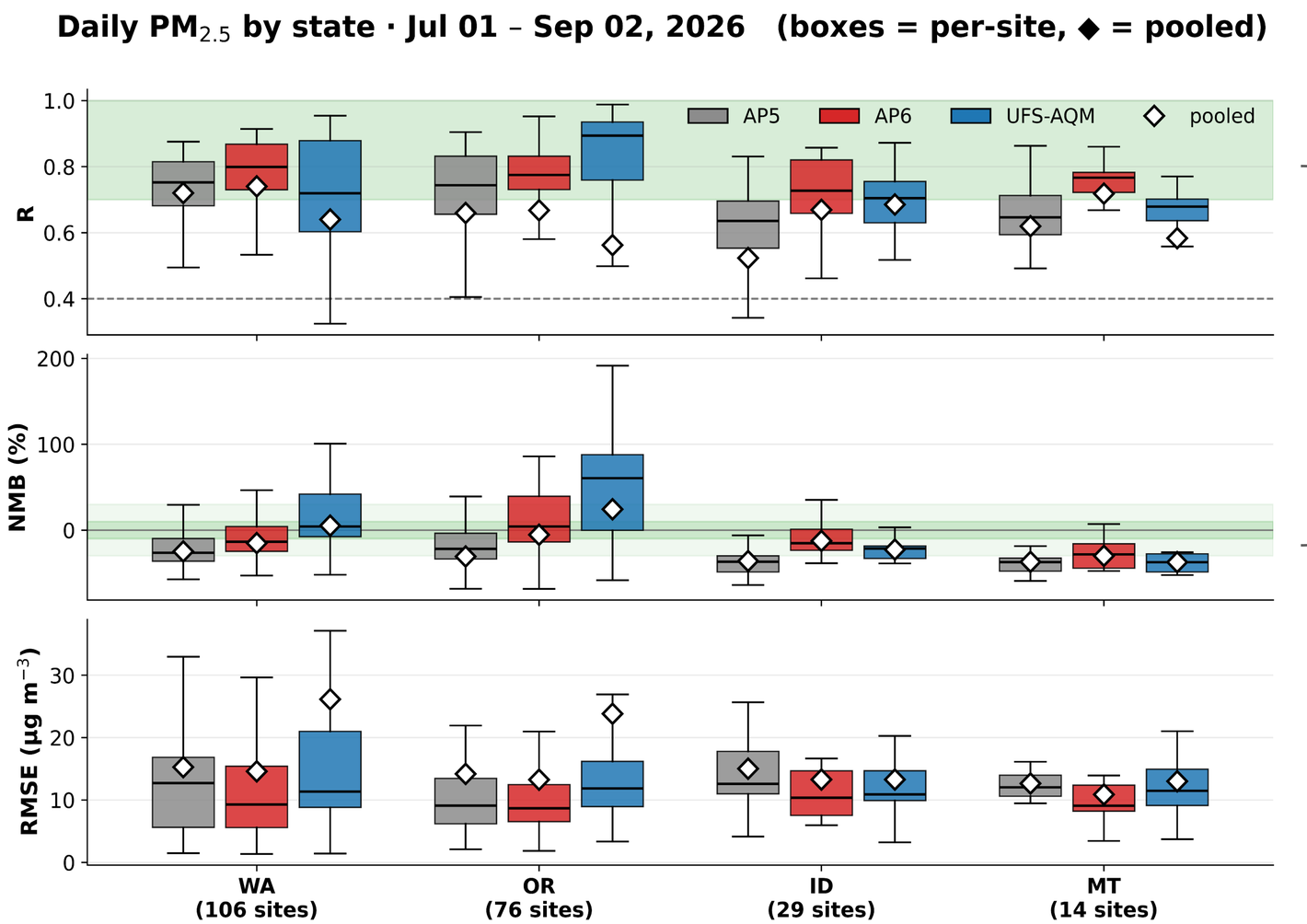
Gridded forecast fields

pm25_20261029.tif GeoTIFF (COG) · 72 bands = hours · ug/m3 · float32, NaN nodata	15.8 MB
pm25_20261029.bin.gz PM2.5 packed binary · uint16 · ug/m3 · scale 10	≤12.1 MB
o3_20261029.tif GeoTIFF (COG) · 72 bands = hours · ppb · float32, NaN nodata	14.0 MB
o3_20261029.bin.gz Ozone packed binary · uint16 · ppb · scale 10	≤12.1 MB
daily_20261029.bin.gz daily products · 24-h PM2.5 mean + ozone MDA8 · 3 day(s)	bin.gz
manifest.json grid, times, scales, bounding box — needed to decode the binaries	JSON

- Monthly Statistic and Monthly emission are delayed; will be ready after we adjust the Anthro. emissions
- AIRPACT-5 operations have ended on the legacy Aeolus HPC system

AIRPACT6 performance evaluation – PM2.5 and O3

Priom's presentation will include more information

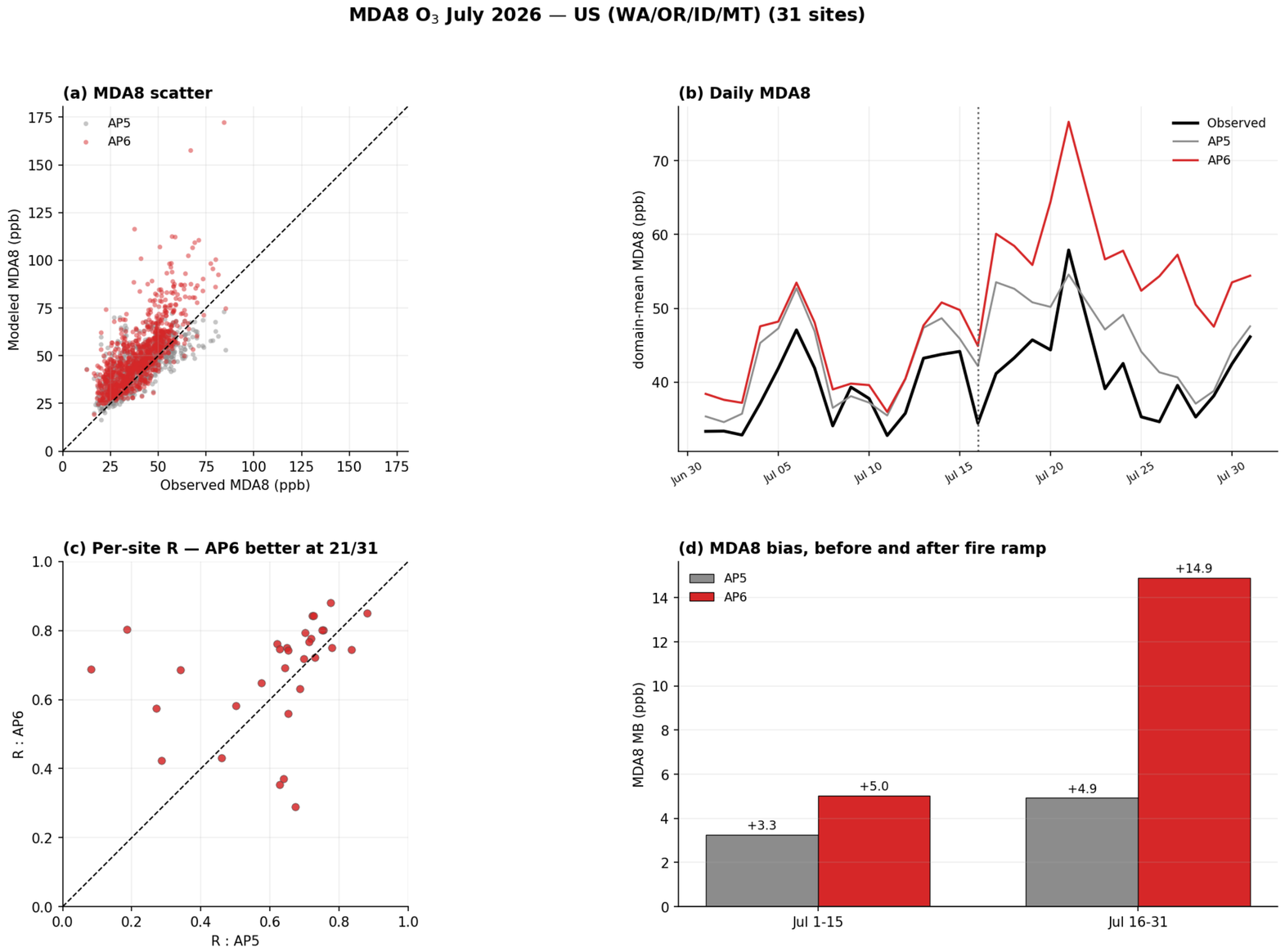


AP6 comply with the benchmark better.

AP6 residual error reduced

(Credits to Priom)

AIRPACT6 performance evaluation – PM2.5 and O3



Priom's presentation will include more information

Our wildfire framework is producing too much Ozone during fire episodes.

(Credits to Priom)

FY27 Proposed Tasks

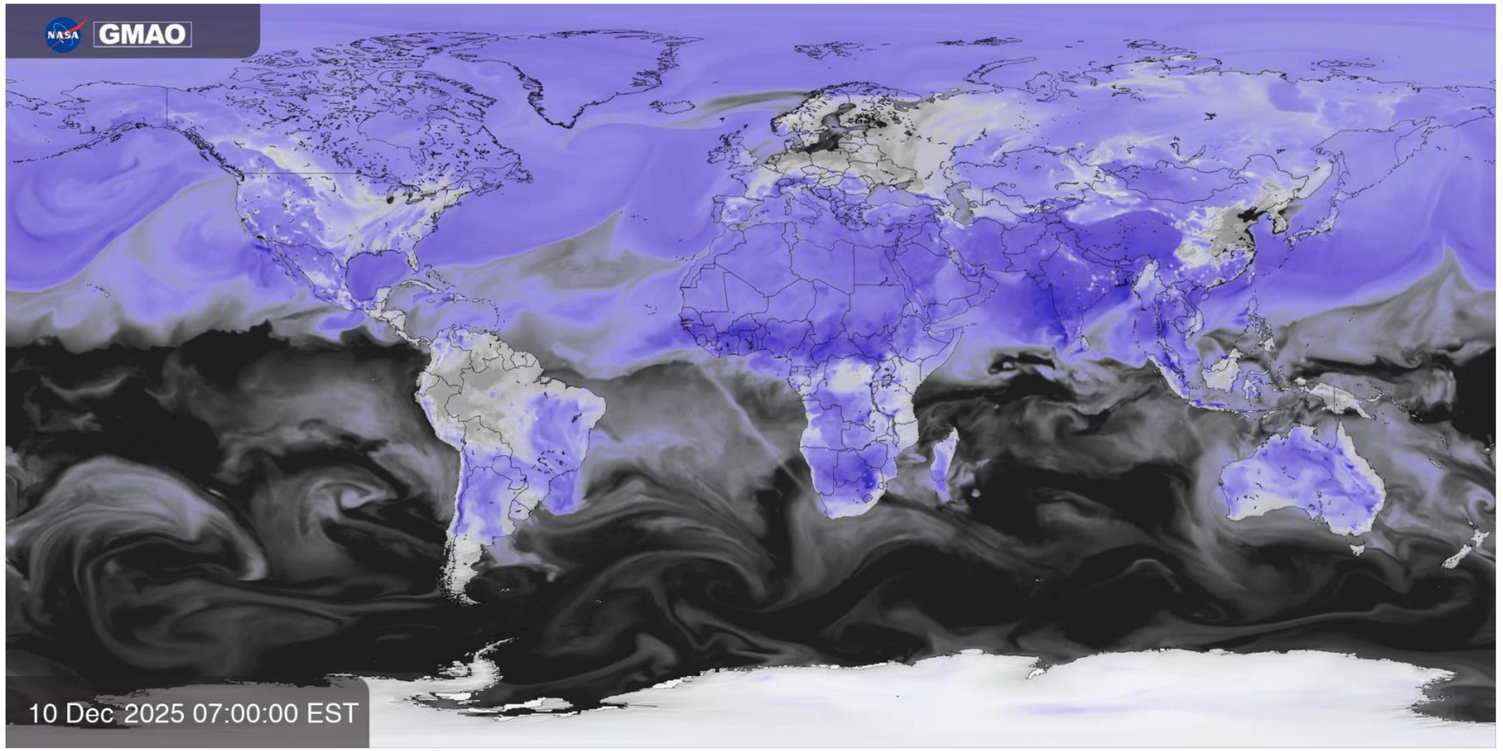
FY27 PROPOSED TASKS

- **Adjust Anthropogenic Emissions – customized 4km surrogates – point source by Dec 31**
- **Display Idaho DEQ's ML forecast on AIRPACT website**
- **Develop a path forward for a hybrid PM2.5 and O3 bias-corrected forecast product, potentially explore NASA TEMPO products**
- **Implement the new chemical mechanism (CRACCM) into AIRPACT6 to potentially improve O3 forecast performance**

Other plans: Implement GEOS-CF as Boundary Condition

GEOS Composition Forecasts

The GEOS Composition Forecasting (GEOS-CF) system produces global, three-dimensional distributions of atmospheric composition, including surface pollutants and tropospheric and stratospheric composition. The GEOS-CF products include a near-real time “analysis” segment, where meteorological fields are informed by other GEOS data assimilation systems, and a 5-day “forecast” segment that projects atmospheric composition using internal model physics and GEOS-Chem chemistry.



NASA GMAO

Advantages

- Higher spatial resolution (~25 km)
- More frequent timesteps (every hour)
- Potential integration of NASA satellite data assimilation in future releases

Summary

- AIRPACT6 in forecast mode is running on WSU HPC Kamiak since June 7 2026
- HMS FRP based wildfire emission inventory is implemented into AIRPACT6
- New interactive map is developed; AIRPACT website improvement is ongoing work
- AIRPACT6 shows better model performance for 2026 wildfire season
- FY27 proposed tasks are underway
- Monthly AIRPACT modeling meeting is ongoing; next meeting will be in November

Thank You!
Any questions?